HOM(F, I_*G') = HOM(I^*F, G') ALONG A CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma spaces-morphisms-lemma-closed-immersion-affine (canonical tag 07U2). cited at proof line 628. Closed immersions are affine.
- Lemma spaces-cohomology-lemma-affine-base-change (canonical tag 07U8). cited at proof line 630. h^*i_*G' = i_{T,*}(h')^*G'.

Source reference. Original lemma 07U2: sources/stacks-chow/tags.txt, source lines 9957–9957. Cited at proof line 628: closed immersions are affine.

Source reference. Original lemma 07U8: sources/stacks-chow/tags.txt, source lines 9963–9963. Cited at proof line 630: h^*i_*G' = i_{T,*}(h')^*G'.

Lemma 0.2. *Let S be a scheme. Let B be an algebraic space over S. Let $i : X' \to X$ be a closed immersion of algebraic spaces over B. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module and let $\mathcal{G}'$ be a quasi-coherent $\mathcal{O}_{X'}$-module. Then*

$$Hom(\mathcal{F}, i_*\mathcal{G}') = Hom(i^*\mathcal{F}, \mathcal{G}')$$

as functors on (Sch/B).

Proof. Let $g : T \to B$ be a morphism where T is a scheme. Denote $i_T : X'_T \to X_T$ the base change of i. Denote $h : X_T \to X$ and $h' : X'_T \to X'$ the projections. Observe that $(h')^*i^*\mathcal{F} = i_T^*h^*\mathcal{F}$. As a closed immersion is affine (Morphisms of Spaces, Lemma 07U2) we have $h^*i_*\mathcal{G} = i_{T,*}(h')^*\mathcal{G}$ by Cohomology of Spaces, Lemma 07U8. Thus we have

$$\begin{aligned} Hom(\mathcal{F}, i_*\mathcal{G}')(T) &= \mathrm{Hom}_{\mathcal{O}_{X_T}}(h^*\mathcal{F}, h^*i_*\mathcal{G}') \\ &= \mathrm{Hom}_{\mathcal{O}_{X_T}}(h^*\mathcal{F}, i_{T,*}(h')^*\mathcal{G}) \\ &= \mathrm{Hom}_{\mathcal{O}_{X'_T}}(i_T^*h^*\mathcal{F}, (h')^*\mathcal{G}) \\ &= \mathrm{Hom}_{\mathcal{O}_{X'_T}}((h')^*i^*\mathcal{F}, (h')^*\mathcal{G}) \\ &= Hom(i^*\mathcal{F}, \mathcal{G}')(T) \end{aligned}$$

as desired. The middle equality follows from the adjointness of the functors $i_{T,*}$ and i_T^*. □

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